

NOMAD: A Node Out-of-Band Module for Attestation and Defense

Department of Cybersecurity, ECE & IoT, Malla Reddy University, Hyderabad

Abstract

Kernel-level runtime defenses face an inherent limit: any detector that runs as software on the host it protects can be blinded or disabled by an attacker who gains sufficient privilege on that same host. Once kernel telemetry, logging, and enforcement all depend on the honesty of the kernel being monitored, a full kernel compromise removes the very evidence needed to detect it. We present NOMAD (Node Out-of-Band Module for Attestation and Defense), a hardware-based attestation and containment module for the Kernel Borderlands runtime defense framework. It runs on a physically separate, minimal computing complex attached to the host, and communicates with the host's software-based watchdog over a dedicated out-of-band link rather than through the monitored operating system. It periodically attests to the host's integrity and, when attestation fails or a containment decision is issued, can act through a hardware relay capable of cutting the host's power independent of the host's own software state. This places the last line of defense outside the failure domain it is meant to defend, so that a full kernel compromise – including rootkits with Ring-0 privilege – does not also compromise the ability to detect and contain it. We describe the design rationale, the threat model it addresses, the two isolated communication planes it exposes, and its current status as an architectural design within the broader Kernel Borderlands project.

Keywords: kernel security, out-of-band attestation, root of trust, hardware watchdog, runtime defense, rootkit resistance

Team Members:

Md Vaseem	2311CS040004	Tejaswini K	2311CS040077
Harshitha B	2311CS040022	Vamshi K	2311CS040078
Sai Raghava D	2311CS040041	Karthik K	2311CS040080
Ashwin Kumar D	2311CS040047	Rupa Yeshvitha K	2311CS040082
Madhumitha J	2311CS040067	Rajesh K	2311CS040084
Jeshwanth K	2311CS040072	Pardhu Sri Rushi Varma K	2311CS040089
Agasthya K	2311CS040075		